The Conditional Partition-Ratio Frontier for State-Weighted $SU(2)$ Control

Abstract

We reduce the remaining one-scale nonlinear problem to a single uniform weighted partition-function inequality. The reduction is exact for every finite compact positive conditional kernel. Two counterexamples show why a local action charge and a pointwise good formula are insufficient: entropy and retained-state normalization must be controlled together.

1 Exact conditional identity

For retained state s, let $w_s(f) > 0$ be the conditional fiber weight and split the compact fiber into $G_s \sqcup B_s$. With next-state map τ, set

$$Z_G(s) = \int_{G_s} w_s(f)\, df, \quad Z_B(s) = \int_{B_s} w_s(f)\, df,$$

$$B_V(s) = \int_{B_s} w_s(f) V(\tau(s,f))\, df.$$

Theorem 1. *The exact weighted bad row of the normalized conditional kernel is*

$$q_V(s) = \frac{B_V(s)}{(Z_G(s) + Z_B(s))V(s)}.$$

If, uniformly in s,

$$Z_G(s) \geq e^{-\ell}\overline{Z}_G(s), \qquad B_V(s) \leq e^{-F}\overline{Z}_G(s)V(s),$$

then $q_V(s) \leq e^{\ell - F}$.

Proof. The identity follows by inserting the normalized kernel $w_s/(Z_G + Z_B)$. For the bound, drop the nonnegative Z_B from the denominator and apply the two hypotheses. □

At the Stage 25 split, the residual ceiling is $r = 0.001215138750921001$ and $\ell = 2.40009685 \cdot 10^{-5}$. Hence it is sufficient to prove

$$F > \ell - \log r = 6.712921011389365.$$

2 Why deterministic charge is insufficient

Give the good channel one microstate of weight one. Give the bad channel $M = \lceil e^{10} \rceil = 22027$ microstates, each with action charge ten and weight e^{-10}. Then

$$Z_B = Me^{-10} = 1.00002425, \qquad \mathbb{P}(B) = 0.50000606.$$

Thus action energy without entropy is not a free-energy estimate.

A second two-state example takes $(Z_G, Z_B) = (1, \varepsilon)$ in one retained state and $(\varepsilon, \varepsilon)$ in another. Pointwise good formulas may be the same, yet the bad probabilities tend to zero and $1/2$. The supremum over positive-mass retained states is essential.

3 First unproved inequality

The constructive target is now explicit: uniformly in finite volume, positive-mass exposure history, retained state, and defect channel, prove

$$\frac{B_V(s)}{\overline{Z}_G(s)V(s)} \le e^{-F_*}, \qquad F_* > 6.71292101139.$$

The finite conditional kernel exists by compactness and positivity, but this uniform free-energy inequality is not established. Therefore one-scale basin entry, RG iteration, continuum construction, and the Hamiltonian mass gap remain open.